Advanced Quantum Mechanics

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Chapter 1

Advanced Quantum Mechanics I: Quantum Dynamics and Symmetries

This chapter rebuilds the basic mathematical spine of the lecture: quantum states as vectors, particles in the position representation, the Hamiltonian-generated unitary time flow, and the symmetry principle that identifies conserved quantities by commuting operators. The goal is to move from abstract postulates to the concrete rules used in practical calculations, with a consistent convention and clear links between infinitesimal generators and finite transformations.

1.1 Review of the quantum kinematic structure

A physical system is represented by vectors in a (complex) vector space. A state is written as a ket $|\psi\rangle$, and its dual by $\langle\psi|$. The inner product $\langle\phi|\psi\rangle$ is a complex number associated to each pair of states.

$$\langle\psi|\psi\rangle = 1 \quad \text{for a normalized state.} \tag{1.1}$$

An observable A is represented by a Hermitian operator $\hat{A}$ ($\hat{A}^\dagger = \hat{A}$). The eigenvalue problem has physical meaning:

$$\hat{A}\,|a\rangle = a\,|a\rangle\,, \qquad a \in \mathbb{R}. \tag{1.2}$$

If the system is in $|a\rangle$, measurement of A is certain to give a. Operational distinguishability is encoded by orthogonality:

$$|\phi\rangle \text{ and } |\psi\rangle \text{ are distinguishable with certainty } \iff \langle\phi|\psi\rangle = 0. \tag{1.3}$$

In finite dimensions these are ordinary vectors; for particles and fields, the basis index becomes continuous and states are represented by functions.

1.2 Particle kinematics in the position representation

For a one-dimensional particle, define position eigenstates $|x\rangle$ by

$$\langle x|x'\rangle = \delta(x - x'). \tag{1.4}$$

The corresponding wavefunction is the position amplitude

$$\psi(x) = \langle x|\psi\rangle. \tag{1.5}$$

Its modulus squared is the position probability density:

$$p(x) = |\psi(x)|^2 = \psi^*(x)\psi(x), \tag{1.6}$$

and for normalizable states $\int dx\, |\psi(x)|^2 = 1$.

The position operator acts by multiplication:

$$(\hat{x}\psi)(x) = x\,\psi(x). \tag{1.7}$$

Distributional caveat. The "vectors" $|x\rangle$ are not ordinary square-integrable functions. Their coordinate overlap is the Dirac delta, and the position eigenfunctions are idealized delta spikes rather than physical packet states. Physical states are square-integrable approximants (e.g., narrow Gaussians).

1.3 Momentum in the same representation

The momentum observable is represented by a differential operator:

$$(\hat{p}\psi)(x) = -i\hbar\,\frac{d\psi(x)}{dx}. \tag{1.8}$$

(If $\hbar = 1$, drop it from formulas.)

Momentum eigenstates satisfy

$$\hat{p}\phi_p(x) = p\,\phi_p(x). \tag{1.9}$$

A convenient representative is

$$\phi_p(x) = \frac{1}{\sqrt{2\pi\hbar}}\,e^{ipx/\hbar}, \tag{1.10}$$

which has $|\phi_p(x)|^2 =$ constant: momentum certainty implies complete delocalization in position. This is the uncertainty tradeoff in precise mathematical form.

1.4 Time evolution as a unitary transformation

Time evolution is represented by a family $U(t)$ of operators mapping each state to a later state:

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle\,, \qquad U(0) = \mathbf{1}. \tag{1.11}$$

A symmetry of information requires inner products to be preserved:

$$\langle\phi|\psi\rangle = \langle U(t)\phi|U(t)\psi\rangle \implies U^\dagger(t)U(t) = \mathbf{1}. \tag{1.12}$$

So $U(t)$ is unitary.

Assuming continuity, for small ϵ,

$$U(\epsilon) = \mathbf{1} - i\epsilon H + \mathcal{O}(\epsilon^2). \tag{1.13}$$

Unitarity to first order gives

$$H^\dagger = H, \tag{1.14}$$

so H is Hermitian and identified as the Hamiltonian.

Now use

$$|\psi(t+\epsilon)\rangle = U(\epsilon)\,|\psi(t)\rangle \tag{1.15}$$

and rearrange:

$$\frac{|\psi(t+\epsilon)\rangle - |\psi(t)\rangle}{\epsilon} = -iH\,|\psi(t)\rangle + \mathcal{O}(\epsilon). \tag{1.16}$$

Hence

$$i\,\frac{d}{dt}\,|\psi(t)\rangle = H\,|\psi(t)\rangle\,. \tag{1.17}$$

For wavefunctions this is

$$i\hbar\,\frac{\partial\psi(x,t)}{\partial t} = H\psi(x,t). \tag{1.18}$$

Finite time is obtained by exponentiation:

$$U(t) = e^{-iHt/\hbar}. \tag{1.19}$$

Stationary states are eigenstates of H:

$$H\,|E_n\rangle = E_n\,|E_n\rangle\,. \tag{1.20}$$

Then

$$|E_n(t)\rangle = e^{-iE_n t/\hbar}\,|E_n\rangle\,, \tag{1.21}$$

so the same Hamiltonian controls both time dependence and allowed energy levels.

1.5 Symmetry as commuting operators

A symmetry V is taken to be a unitary map on states that preserves all physical distinctions. If it represents a true symmetry of dynamics, then evolving then transforming is equivalent to transforming then evolving:

$$V\,U(t) = U(t)\,V, \qquad \forall t. \tag{1.22}$$

Using $U(t) = e^{-iHt/\hbar}$, this is equivalent to

$$[V, H] = 0. \tag{1.23}$$

For a continuous family $V(\epsilon) = \mathbf{1} - i\epsilon G + \mathcal{O}(\epsilon^2)$, the same logic gives

$$[G, H] = 0. \tag{1.24}$$

When $G = G^\dagger$, this is exactly a conserved generator:

$$\frac{d}{dt}\langle G\rangle = \frac{i}{\hbar}\langle [H, G]\rangle = 0. \tag{1.25}$$

Thus symmetry $\leftrightarrow$ conserved quantity is the operator-level statement.

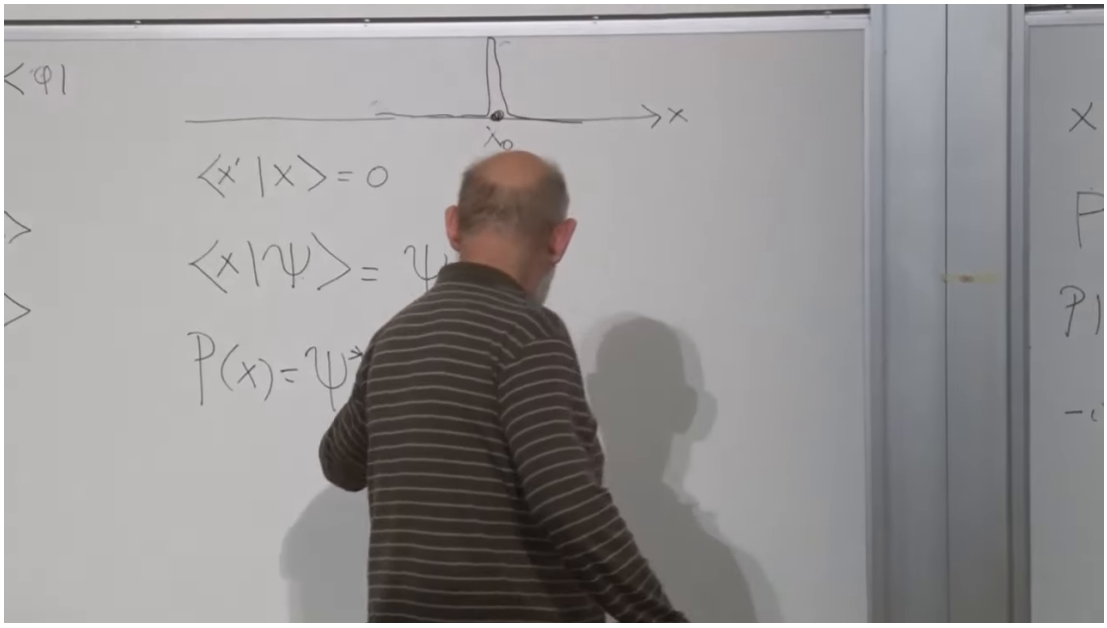


Figure 1.1: Infinitesimal-to-finite evolution structure for $U(\epsilon)$.

$$
\begin{tikzcd}
|\psi_1\rangle \arrow[r, "U(t)"] \arrow[d, "V"'] & |\psi_2\rangle = U(t)\,|\psi_1\rangle \arrow[d, "V"] \\
V\,|\psi_1\rangle \arrow[r, "U(t)"] & V\,|\psi_2\rangle = U(t)\,V\,|\psi_1\rangle
\end{tikzcd}
$$

Figure 1.2: A symmetry operation V commutes with time evolution $U(t)$ if the two routes between states coincide.

1.6 Continuous symmetry example: spatial translations

Define the spatial translation operator $T(\epsilon)$ acting on wavefunctions by

$$(T(\epsilon)\psi)(x) = \psi(x+\epsilon), \tag{1.26}$$

which is a rightward shift by ϵ. For small ϵ,

$$\psi(x+\epsilon) = \psi(x) + \epsilon\,\partial_x\psi(x) + \mathcal{O}(\epsilon^2), \tag{1.27}$$

so

$$T(\epsilon) = \mathbf{1} + \epsilon\,\partial_x + \mathcal{O}(\epsilon^2) = \mathbf{1} - \frac{i\epsilon}{\hbar}\,\hat{p}_x + \mathcal{O}(\epsilon^2). \tag{1.28}$$

Hence

$$T(\epsilon) = \exp\left(-\frac{i}{\hbar}\epsilon\,\hat{p}_x\right), \tag{1.29}$$

so the generator is

$$G_{\text{trans}} = \frac{\hat{p}_x}{\hbar}. \tag{1.30}$$

For a free particle,

$$H = \frac{\hat{p}_x^2}{2m}, \tag{1.31}$$

and $[H, \hat{p}_x] = 0$, so translation is a symmetry and $\hat{p}_x$ is conserved. Thus the same operator appears as both "generator of translations" and "conserved quantity," illustrating the lecture's central symmetry–conservation link.

1.7 Summary

The lecture's core chain is: state vectors → continuous basis amplitudes → operator representation → unitary time flow. Continuity plus unitarity forces $U(t)$ to be generated by a Hermitian Hamiltonian and yields the Schrödinger equations. A symmetry is an additional unitary map compatible with $U(t)$, which becomes a commutator condition with H. For continuous symmetries, infinitesimal generators G define finite transformations via exponentials, and commuting generators are conserved quantities. Translations provide the prototype: $T(\epsilon) = e^{-i\epsilon\hat{p}_x/\hbar}$, and for $H = p^2/2m$, momentum conservation follows directly from $[H, \hat{p}_x] = 0$.

Chapter 2

Symmetry, Degeneracy, and Angular Momentum for Quantum Rotational Systems

This chapter develops a symmetry-first view of rotational quantum systems. We define symmetries as operators commuting with the Hamiltonian, distinguish generic level coincidences from symmetry-protected multiplets, and build angular-momentum structure from the commutator algebra of generators. The particle on a circle provides an explicit worked model where quantized m-values, reflection pairing, and ladder operators emerge from operator algebra rather than ad hoc assumptions.

2.1 Symmetry as an invariance principle

A symmetry is an operation that leaves the quantum description unchanged. For a unitary transformation S, invariance of dynamics is the condition

$$SHS^\dagger = H$$

equivalently

$$[H, S] = 0.$$

Because S commutes with H, Schrödinger evolution is covariant under S: if $|\psi(t)\rangle$ evolves by H, then $S|\psi(t)\rangle$ evolves in the same way. In particular, E-eigenvalues are mapped to the same E-eigenspaces.

Symmetries can be exact (invariance of the full Hamiltonian), approximate (effective in a regime), or accidental (valid only for very specific parameter choices). Our focus is on exact symmetries because they control spectral structure.

2.2 Symmetry examples and the organizing role of operators

The same formal rule $[H, S] = 0$ unifies many familiar examples:

- spatial translations and rotations,

- interchange of identical particles,
- relabeling symmetries of a periodic lattice (discrete translation in crystal index),
- discrete mirror or inversion operations.

In practice, this is less about geometry alone and more about algebraic leverage: a set of commuting operators gives you a good quantum-number basis, while noncommuting families often generate nontrivial multiplets. Symmetry language gives a compact way to classify allowed degeneracies before solving differential equations.

Figure 2.1: Prototype symmetry motifs: geometric and discrete transformations that can commute with the Hamiltonian.

2.3 Degeneracy, symmetry, and when degeneracy is protected

For a bounded discrete spectrum, exact equality of two nearby energies is rare unless there is structure behind it. We call this *accidental* coincidence when no symmetry explanation exists. In contrast, symmetry-protected degeneracies are stable under small perturbations that preserve the same symmetry.

Proposition 2.1. *Let A and B be Hermitian symmetry generators commuting with H: $[A, H] = [B, H] = 0$. Define*

$$C := \frac{1}{i\hbar}[A, B].$$

Then C is Hermitian and $[C, H] = 0$.

Proof. Since A and B are Hermitian, $[A, B]^\dagger = -[A, B]$, so C is Hermitian. Then

$$[C, H] = \frac{1}{i\hbar}([AB, H] - [BA, H])$$

and expanding each commutator gives zero because both A and B commute with H. Hence $[C, H] = 0$. $\square$

Thus one can grow a closed set of symmetry generators by commutators. If that set closes to a Lie algebra with nontrivial structure constants, the representation theory is what drives degeneracy and spectral multiplets.

2.4 Particle on a circle: quantized L_z from single-valued wave functions

Let $\theta \in [0, 2\pi)$ label the position of a particle constrained to a circle. A small rotation by ϵ acts as

$$(\mathcal{R}(\epsilon)\psi)(\theta) = \psi(\theta - \epsilon).$$

Its infinitesimal form defines the generator L_z:

$$\mathcal{R}(\epsilon)\psi(\theta) \approx \left(1 - \frac{i\epsilon}{\hbar} L_z\right)\psi(\theta), \qquad L_z = -i\hbar\frac{d}{d\theta}.$$

Hence

$$L_z \left|\psi_m\right\rangle = m\hbar \left|\psi_m\right\rangle \quad \Longleftrightarrow \quad -i\hbar\frac{d\psi_m}{d\theta} = m\hbar\psi_m$$

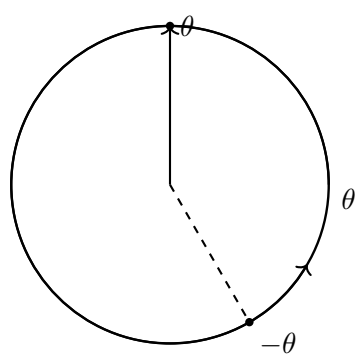
and therefore

$$\psi_m(\theta) = \frac{1}{\sqrt{2\pi}} e^{im\theta}.$$

Single-valuedness on $0 \sim 2\pi$ gives

$$\psi_m(\theta + 2\pi) = \psi_m(\theta) \iff e^{i2\pi m} = 1 \iff m \in \mathbb{Z}.$$

So the orbital angular-momentum quantum numbers are $m = 0, \pm 1, \pm 2, \ldots$.



Identification of θ and $\theta + 2\pi$: $\psi(\theta + 2\pi) = \psi(\theta)$

Figure 2.2: Circle coordinate, opposite angles, and periodicity condition.

2.5 Mirror symmetry and the $m \leftrightarrow -m$ pairing

Introduce a mirror (reflection) operator $\mathcal{M}$ with

$$(\mathcal{M}\psi)(\theta) = \psi(-\theta).$$

Its action on a momentum eigenmode is

$$\mathcal{M}\psi_m(\theta) = \psi_{-m}(\theta).$$

On a circle eigenstate:

$$\mathcal{M}L_z\psi_m = m\hbar\,\mathcal{M}\psi_m, \qquad L_z\mathcal{M}\psi_m = -m\hbar\,\mathcal{M}\psi_m,$$

so

$$[\mathcal{M}, L_z]\psi_m = 2m\hbar\,\mathcal{M}\psi_m \neq 0 \quad (m \neq 0).$$

Thus $\mathcal{M}$ does not commute with L_z, even though both may commute with H in a symmetric setup. When both symmetries are present (no mirror-breaking field), states m and $-m$ are paired at equal energy.

Important warning from the formalism: equal probabilities $|\psi_m(\theta)|^2 = |\psi_{-m}(\theta)|^2$ does not imply identical wave functions. Degeneracy concerns states and operators, not only densities.

A uniform magnetic field is a standard counterexample: it can split $\pm m$ levels while preserving some rotational structure of orbital motion in other settings. Symmetry arguments apply to what the Hamiltonian does, not to geometric intuition alone.

2.6 Generators, commutator algebra, and closure

For continuous symmetries a small transformation is written as

$$U(\epsilon) = 1 - \frac{i\epsilon}{\hbar}G + \mathcal{O}(\epsilon^2),$$

with Hermitian generator G. The commutator of two generators is another generator of another direction in the same symmetry structure if closure holds.

A useful worked pattern is:

$$[\,\mathcal{M}, L_z\,] \neq 0, \qquad \text{while } \; [H, L_z] = 0, \; [H, \mathcal{M}] = 0 \text{ (if mirror symmetric)}.$$

Hence $\mathcal{M}$ and L_z together generate two operations whose noncommutativity forces additional structure.

2.7 Three-dimensional angular momentum and ladder structure

For a particle in 3D,

$$L_x = yp_z - zp_y, \qquad L_y = zp_x - xp_z, \qquad L_z = xp_y - yp_x,$$

with canonical $[x_i, p_j] = i\hbar\delta_{ij}$. A compact expansion gives

$$[L_x, L_y] = i\hbar L_z,$$

and cyclic permutations. Equivalently,

$$[L_i, L_j] = i\hbar \sum_k \epsilon_{ijk} L_k.$$

For a rotationally invariant Hamiltonian, $[H, L_i] = 0$, so L_x, L_y, L_z are conserved quantities.

Define raising and lowering operators

$$L_\pm := L_x \pm iL_y.$$

Using the algebra above,

$$[L_z, L_\pm] = \pm\hbar L_\pm, \qquad [L_+, L_-] = 2\hbar L_z.$$

If $|\ell, m\rangle$ is an eigenvector of L_z,

$$L_z|\ell, m\rangle = m\hbar|\ell, m\rangle,$$

then

$$L_z(L_\pm|\ell, m\rangle) = (m \pm 1)\hbar(L_\pm|\ell, m\rangle),$$

so $L_\pm$ create a state with adjacent m-value when nonzero.

In a finite ladder representation, the chain terminates:

$$L_+|\ell, \ell\rangle = 0, \qquad L_-|\ell, -\ell\rangle = 0,$$

giving $m = -\ell, -\ell+1, \ldots, \ell$, hence $2\ell+1$ states in one rotational multiplet. For orbital wave functions on configuration space, $\ell, m \in \mathbb{Z}$ and $m \in \{-\ell, \ldots, \ell\}$. For more general angular-momentum types (including spin), half-integer m spectra are also possible; in this lecture the orbital circle argument selects the integer branch.

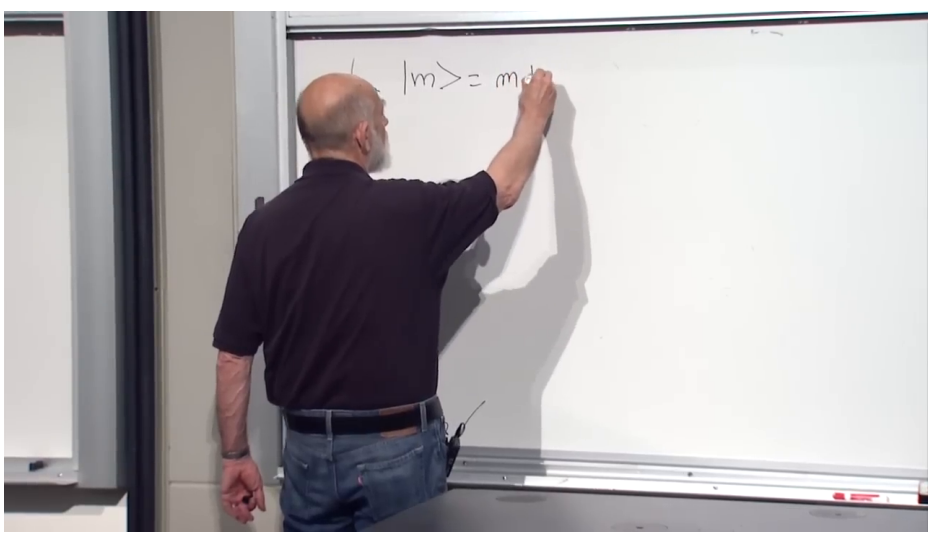


Figure 2.3: Illustrative ladder picture: $L_\pm$ connect adjacent m-levels inside a symmetry multiplet.

2.8 Summary

Symmetry enters as a commutation criterion with the Hamiltonian. Single symmetries can explain invariant spectral structure, but genuine multiplets require a noncommuting symmetry set whose commutator algebra closes. The circle provides the cleanest demonstration: rotational invariance gives $L_z = -i\hbar\partial_\theta$, single-valuedness gives integer m, and mirror symmetry links m to $-m$. In three dimensions, the algebra $[L_i, L_j] = i\hbar\epsilon_{ijk}L_k$ yields ladder operators $L_\pm$, a finite m-tower for each ℓ, and equal-energy multiplets in rotationally invariant systems.

Chapter 3

Angular Momentum, Central Forces, and the Oscillator Ladder

These notes follow Leonard Susskind's *Advanced Quantum Mechanics* Lecture 3, curated by LazyingArt LLC. The lecture opens as a continuation of the previous discussion of angular momentum, but almost immediately shifts from algebra to physics: angular momentum is the language of how a wavefunction depends on angles. From there the hour unfolds in two acts. First, angular-momentum multiplets are reconstructed and then used to reduce the central-force Schrödinger equation to a one-dimensional radial problem. Second, the same ladder-operator instinct reappears in the harmonic oscillator. Throughout, Susskind suppresses factors of $\hbar$ on the board; they are restored here.

3.1 Angular Momentum as Angular Dependence of the Wavefunction

Susskind begins by recalling the angular-momentum operators and the multiplets they form, but he does not stay in recap mode for long. Instead he asks what angular momentum is really about. His example is a particle moving in a central force field. Classically, the angular momentum

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \tag{3.1}$$

is conserved, so both its magnitude and its direction remain fixed. The orbit therefore stays in a fixed plane perpendicular to $\mathbf{L}$.

Quantum mechanically the particle is described by a wavefunction, and the point is no longer the orbit itself but the way the wavefunction changes under rotation. In a central-force problem the wavefunction may be written as a function of radius and angles,

$$\Psi = \Psi(r, \theta, \phi), \tag{3.2}$$

and angular momentum acts only on the angular part.

A two-dimensional prototype. Susskind briefly returns to two dimensions because there the basic idea is completely transparent. If motion is confined to a plane, the only relevant angular

momentum is the generator of rotations about the perpendicular axis:

$$L_z = -i\hbar \frac{\partial}{\partial \theta}. \tag{3.3}$$

An angular-momentum eigenstate therefore satisfies

$$L_z \psi(r,\theta) = \hbar \ell \, \psi(r,\theta), \tag{3.4}$$

or

$$-i\hbar \frac{\partial}{\partial \theta} \psi(r,\theta) = \hbar \ell \, \psi(r,\theta). \tag{3.5}$$

Here the derivative is only with respect to θ. The radial coordinate is a spectator. Solving for the θ-dependence gives

$$\psi(r,\theta) = \chi(r) e^{i\ell\theta}, \tag{3.6}$$

where ℓ must be an integer so that the wavefunction returns to itself when θ goes around the circle by 2π.

This is the first structural lesson of the lecture. Angular momentum is determined by the angular dependence of the wavefunction. The radial factor $\chi(r)$ may matter for the energy, but it does not matter for the angular-momentum eigenvalue.

Back to three dimensions. The same statement survives in three dimensions. One writes the wavefunction schematically as

$$\Psi(r,\theta,\phi) = R(r)\, Y_{\ell m}(\theta,\phi), \tag{3.7}$$

where the angular factor $Y_{\ell m}$ carries the rotational quantum numbers. These are the spherical harmonics. Susskind's emphasis is that one need not know their explicit formulas at this stage. What matters is that the angular problem and the radial problem separate, and angular momentum belongs entirely to the first of them.

With that physical point fixed, he returns to the operator algebra.

3.2 From the L_z Basis to Finite Multiplets

The next move, as Susskind says in a distinctly Dirac-like spirit, is to pick one component of angular momentum and work in its eigenbasis. Take L_z. The basic commutators are

$$[L_x, L_y] = i\hbar L_z, \qquad [L_y, L_z] = i\hbar L_x, \qquad [L_z, L_x] = i\hbar L_y. \tag{3.8}$$

Now define

$$L_\pm = L_x \pm i L_y. \tag{3.9}$$

A short computation gives

$$[L_z, L_\pm] = \pm\hbar L_\pm. \tag{3.10}$$

Therefore, if

$$L_z|\ell, m\rangle = \hbar m \, |\ell, m\rangle, \tag{3.11}$$

then

$$L_z\big(L_\pm|\ell,m\rangle\big) = \big(L_\pm L_z + [L_z, L_\pm]\big)|\ell,m\rangle \tag{3.12}$$

$$= \hbar(m \pm 1)\big(L_\pm|\ell,m\rangle\big). \tag{3.13}$$

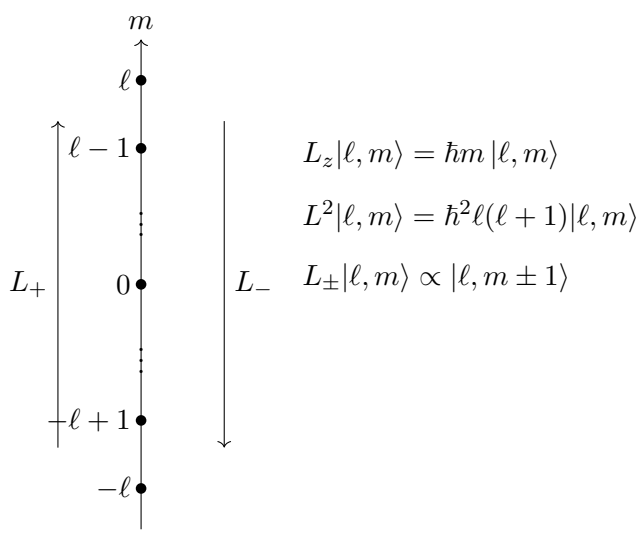


Figure 3.1: The finite angular-momentum ladder suggested by the blackboard discussion. The ladder operators move in unit steps of m, and the multiplet terminates symmetrically at $m = \pm\ell$.

So L_+ raises m by one unit and L_- lowers it by one unit:

$$L_\pm|\ell, m\rangle \propto |\ell, m \pm 1\rangle. \tag{3.14}$$

That is the first algebraic payoff. The second is the question Susskind immediately asks: does the ladder continue forever? It can only stop if eventually one of the raised or lowered states vanishes:

$$L_+|\text{top}\rangle = 0, \qquad L_-|\text{bottom}\rangle = 0. \tag{3.15}$$

Rotational symmetry then forces the spectrum to be symmetric about zero. There is nothing fundamentally privileged about the $+z$-direction relative to the $-z$-direction. Hence a finite ladder must run from some highest value $m = \ell$ down to the reflected value $m = -\ell$:

$$m = -\ell, -\ell + 1, \ldots, \ell - 1, \ell. \tag{3.16}$$

The number of states is therefore

$$2\ell + 1. \tag{3.17}$$

Susskind makes one further observation here. If the steps are always in integers, then a finite symmetric ladder can begin either at an integer or at a half-integer. Thus the abstract angular-momentum algebra allows both integer and half-integer multiplets. Later, when the discussion specializes to orbital wavefunctions on the sphere, only integer ℓ remain.

Remark 3.1. The lecture briefly mentions the possibility of infinite ladders. They are mathematically conceivable, but they are set aside as physically pathological for the finite-energy situations of interest here.

3.3 What a Multiplet Has in Common

Once the ladder has been built, Susskind asks what characterizes the entire multiplet. The answer is the square of the total angular momentum,

$$L^2 = L_x^2 + L_y^2 + L_z^2. \tag{3.18}$$

To evaluate it, he rewrites $L_x^2 + L_y^2$ in terms of the ladder operators. The ordered product is

$$\begin{aligned} L_- L_+ &= (L_x - iL_y)(L_x + iL_y) && (3.19)\\ &= L_x^2 + L_y^2 + iL_xL_y - iL_yL_x && (3.20)\\ &= L_x^2 + L_y^2 + i[L_x, L_y] && (3.21)\\ &= L_x^2 + L_y^2 - \hbar L_z. && (3.22) \end{aligned}$$

Therefore

$$L^2 = L_z^2 + \hbar L_z + L_- L_+. \tag{3.23}$$

There is also the equally valid alternative

$$L^2 = L_z^2 - \hbar L_z + L_+ L_-, \tag{3.24}$$

which corresponds to factoring in the opposite order. An audience member asks precisely this question in the lecture, and Susskind's answer is that the same derivation works if one applies the reordered identity to the bottom state instead of the top.

Worked derivation of $L^2 = \hbar^2 \ell(\ell+1)$. Apply the first ordered form to the highest state of the multiplet:

$$\begin{aligned} L^2|\ell,\ell\rangle &= \left(L_z^2 + \hbar L_z + L_- L_+\right)|\ell,\ell\rangle && (3.25)\\ &= \left(\hbar^2\ell^2 + \hbar^2\ell\right)|\ell,\ell\rangle + L_-(L_+|\ell,\ell\rangle). && (3.26) \end{aligned}$$

By definition of the top state,

$$L_+|\ell,\ell\rangle = 0, \tag{3.27}$$

and so

$$L^2|\ell,\ell\rangle = \hbar^2\ell(\ell+1)|\ell,\ell\rangle. \tag{3.28}$$

This is the blackboard formula $L^2 = \ell(\ell+1)$ with the suppressed factor of $\hbar^2$ restored.

The lecture then pushes one step further. This is not just a property of the top state. In fact,

$$[L^2, L_i] = 0, \qquad i = x, y, z, \tag{3.29}$$

and therefore L^2 commutes with $L_\pm$ as well. For example,

$$\begin{aligned} L^2(L_-|\ell,\ell\rangle) &= L_-(L^2|\ell,\ell\rangle) && (3.30)\\ &= \hbar^2\ell(\ell+1)(L_-|\ell,\ell\rangle). && (3.31) \end{aligned}$$

So the next state down has the same L^2-eigenvalue, and the same argument propagates all the way to the bottom of the ladder.

This is the meaning of the multiplet. All $2\ell+1$ states have the same total angular momentum, but different projections on the z-axis:

$$L^2|\ell,m\rangle = \hbar^2\ell(\ell+1)|\ell,m\rangle, \qquad L_z|\ell,m\rangle = \hbar m\,|\ell,m\rangle. \tag{3.32}$$

In the orbital case these states are represented by the spherical harmonics $Y_{\ell m}(\theta,\phi)$, with $\ell = 0, 1, 2, \ldots$ and $m = -\ell, \ldots, \ell$.

Susskind then makes the transition that matters for atomic physics. In a central-force problem the Hamiltonian commutes with angular momentum, so all members of a fixed ℓ-multiplet have the same energy:

$$[H, L_i] = 0 \qquad \text{for } V = V(r). \tag{3.33}$$

That closes the angular-momentum recap and opens the central-force problem.

3.4 From Three Dimensions to an Effective Radial Problem

At this point Susskind says explicitly that he will use a little classical intuition, even a tiny bit of cheating, to motivate the reduction of the central-force problem. The starting Hamiltonian is

$$H = \frac{p^2}{2m} + V(r), \tag{3.34}$$

with p^2 the full three-dimensional momentum squared and $V(r)$ depending only on the distance from the origin.

Classical reduction. Because the force is central, angular momentum is conserved. Classically one may therefore choose coordinates so that the orbit lies in the xy-plane. In that plane decompose the momentum into a radial component p_r and an angular component p_θ:

$$p^2 = p_r^2 + p_\theta^2. \tag{3.35}$$

The Hamiltonian becomes

$$H = \frac{p_r^2 + p_\theta^2}{2m} + V(r). \tag{3.36}$$

Now use the magnitude of the classical angular momentum:

$$L = r\, p_\theta. \tag{3.37}$$

Hence

$$L^2 = r^2 p_\theta^2, \qquad p_\theta^2 = \frac{L^2}{r^2}. \tag{3.38}$$

Substituting back gives the board equation

$$H = \frac{p_r^2}{2m} + \frac{L^2}{2mr^2} + V(r). \tag{3.39}$$

This is the effective radial Hamiltonian. Once the angular momentum is fixed, everything angular has been frozen into the parameter L, and the remaining motion is one-dimensional in r.

It is natural to define the effective potential

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} + V(r). \tag{3.40}$$

The extra $1/r^2$ term is the centrifugal barrier. It is repulsive, not attractive: as r decreases, the potential rises sharply, so the corresponding force pushes the particle away from the origin.

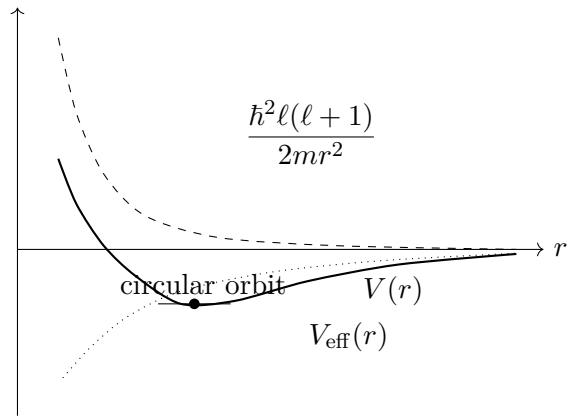


Figure 3.2: The effective radial potential for an attractive central force. The Coulomb tail $V(r) \sim -1/r$ and the centrifugal barrier $\sim 1/r^2$ combine into a one-dimensional potential with a minimum when $\ell \neq 0$.

What the effective potential means. For a Coulomb attraction $V(r) \sim -1/r$, the barrier behaves as $+1/r^2$. Near the origin the $1/r^2$ term dominates; far away the $1/r$ term dominates. Their sum has a minimum. A particle sitting at that minimum executes a circular orbit, while a particle with slightly different energy oscillates radially between a minimum and maximum radius. In the Coulomb case those radial oscillations are the classical elliptical orbits.

There is one exceptional case, emphasized later in the discussion: if $\ell = 0$, then the centrifugal barrier vanishes. Classically nothing prevents the motion from heading straight through the origin.

3.5 The Quantum Radial Equation

Susskind now turns to the quantum problem, but he keeps the classical reduction in view. He does not carry out every angular derivative on the board. Instead he uses the classical form of the Hamiltonian as a guide and then states the separated quantum equation. The omitted manipulations are real, but they do not change the basic story.

The full time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\Psi + V(r)\Psi = E\Psi. \tag{3.41}$$

Now choose a state of definite angular momentum:

$$\Psi_{\ell m}(r,\theta,\phi) = \psi(r)\, Y_{\ell m}(\theta,\phi). \tag{3.42}$$

Once ℓ and m are fixed, the angular dependence has already been solved by the spherical harmonic. The remaining unknown is the radial function $\psi(r)$.

The radial momentum operator is

$$p_r = -i\hbar\,\frac{d}{dr}, \tag{3.43}$$

so that

$$p_r^2 \to -\hbar^2\,\frac{d^2}{dr^2}. \tag{3.44}$$

Susskind is slightly casual about the sign when he speaks it aloud, but only p_r^2 appears in the Hamiltonian, so the final radial kinetic term is unambiguous.

On a state of definite ℓ, one replaces L^2 by $\hbar^2\ell(\ell+1)$. Thus the radial Hamiltonian acts as

$$H\psi(r) = E\psi(r), \tag{3.45}$$

with

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dr^2} + \frac{\hbar^2\ell(\ell+1)}{2mr^2} + V(r), \tag{3.46}$$

and therefore

$$-\frac{\hbar^2}{2m}\frac{d^2}{dr^2}\psi(r) + \frac{\hbar^2\ell(\ell+1)}{2mr^2}\psi(r) + V(r)\psi(r) = E\psi(r). \tag{3.47}$$

Remark 3.2. The lecture writes this as a pure second-order equation for $\psi(r)$. In standard separation of variables this clean one-dimensional form is the equation for the reduced radial function $u_\ell(r) = rR_\ell(r)$. To stay close to Susskind's notation while remaining precise, it is best to read $\psi(r)$ here as that reduced radial variable.

This is the main payoff of the angular-momentum recap. The original three-dimensional partial differential equation has become an ordinary differential equation in one variable. The angular structure has not disappeared; it has been compressed into the single parameter ℓ.

There is also a conceptual point that surfaces in the question period. Quantum mechanically one cannot literally rotate the whole wavefunction into a fixed plane, as one does classically for a single orbit. The memory of full three-dimensional motion survives in the fact that the relevant operator is L^2, not L_z^2. If one could replace L^2 by L_z^2, one would get something like ℓ^2. Instead one gets the genuinely three-dimensional result $\ell(\ell+1)$.

Susskind stops short of solving the radial equation explicitly. That, he says, is a matter of differential equations, and the lecture is after the structure of the spectrum rather than the explicit special functions.

3.6 Nodes, Degeneracies, and the Coulomb Accident

Having reduced the problem to one dimension, Susskind turns immediately to what one can say about the spectrum without solving the equation in detail.

Node counting. For fixed ℓ, the radial Schrödinger equation is a one-dimensional bound-state problem. The lowest bound state has no nodes, the first excited state has one node, the next has two, and so on. Here a node means a zero of the radial wavefunction. Susskind gives the physical reason in his usual way: more nodes mean more rapid wiggles; more rapid wiggles mean larger momentum; larger momentum means higher kinetic energy. In this sense the node count organizes the bound-state spectrum.

The generic central-force spectrum. Each value of ℓ gives a different radial equation because the centrifugal term depends on ℓ. Thus one obtains a separate tower of radial states for each ℓ. The $\ell = 0$ tower sits lowest because the centrifugal barrier is absent there. The $\ell = 1$ tower is shifted upward, the $\ell = 2$ tower upward again, and so on. At every level in the ℓ-th tower there are

$$2\ell + 1 \tag{3.48}$$

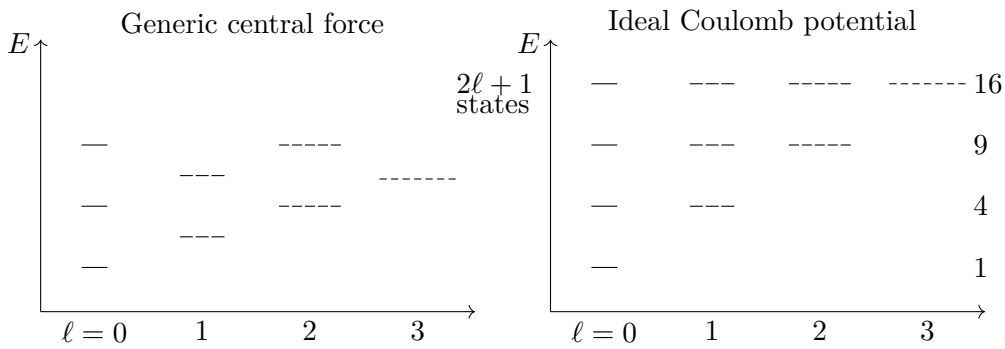


Figure 3.3: Left: the generic qualitative spectrum of an attractive central force. Each value of ℓ has its own radial tower, shifted upward by the centrifugal term, and each level carries degeneracy $2\ell + 1$. Right: in the ideal Coulomb problem, levels from different ℓ-towers align into shells with orbital degeneracies $1, 4, 9, 16, \ldots$.

degenerate states, corresponding to the allowed values of m.

At sufficiently high energy the states cease to be bound. Classically the particle escapes; quantum mechanically the discrete spectrum gives way to the continuum. For the bound states of an attractive potential, the energies lie below that escape threshold.

The special case $\ell = 0$. One audience question returns to the missing centrifugal barrier when $\ell = 0$. Classically, if the potential is Coulombic, the particle would simply run toward the center. Quantum mechanically that does not happen. The lowest state cannot be both sharply localized at the origin and at rest there. The uncertainty principle forces a compromise: the ground-state wavefunction spreads over a finite region, and the corresponding energy is finite rather than negatively infinite.

The Coulomb accident. For the special potential $V(r) \propto -1/r$, Susskind describes a remarkable extra degeneracy. The lowest $\ell = 1$ state has exactly the same energy as the first excited $\ell = 0$ state. The lowest $\ell = 2$ state has the same energy as the second excited $\ell = 0$ state. More generally, the lowest state in the ℓ-tower aligns with the ℓ-node state in the $\ell = 0$ tower. The result is a sequence of shells whose orbital degeneracies are

$$1, \qquad 4, \qquad 9, \qquad 16, \qquad \ldots \tag{3.49}$$

or, equivalently,

$$\sum_{k=0}^{n-1} (2k + 1) = n^2. \tag{3.50}$$

Susskind repeatedly warns that this is special to the ideal nonrelativistic Coulomb problem. A finite nuclear size, relativistic corrections, and spin all spoil it. The hydrogen atom approximately displays the pattern, but not exactly. In particular, because spin has not yet been included, the analysis is missing a factor of two. Real atomic shell counts begin

$$2, \qquad 8, \qquad 18, \qquad 32, \qquad \ldots \tag{3.51}$$

rather than

$$1, \qquad 4, \qquad 9, \qquad 16, \qquad \ldots \tag{3.52}$$

The lecture also pauses on a subtle question about wavefunctions. Bound states do not have compact support. They do not literally vanish beyond some finite radius. Instead they fall off rapidly, typically exponentially, so that the probability of finding the particle very far away is nonzero but extremely small.

That completes the atomic part of the lecture. Susskind then puts atoms aside for the moment and turns to a new system in which the same algebraic luck appears again.

3.7 The Harmonic Oscillator: Dirac's Trick Again

The transition to the harmonic oscillator is not a digression so much as a replay. Susskind emphasizes that many small oscillations in physics are approximately harmonic, and he even notes that oscillations near the bottom of an effective radial well are an example. The oscillator is the second case in which ladder operators solve the spectrum almost at once.

Take the mass to be 1 and write the spring constant as $k = \omega^2$. Then the classical Hamiltonian is

$$H = \frac{p^2}{2} + \frac{\omega^2 x^2}{2}. \tag{3.53}$$

Hamilton's equations are

$$\dot{x} = \frac{\partial H}{\partial p} = p, \qquad \dot{p} = -\frac{\partial H}{\partial x} = -\omega^2 x. \tag{3.54}$$

Therefore

$$\ddot{x} = -\omega^2 x, \tag{3.55}$$

which identifies ω as the classical frequency of oscillation.

Factorization and the zero-point term. Now comes the Dirac-like move. Since the Hamiltonian is a sum of squares, try to factor it:

$$\frac{1}{2}(p + i\omega x)(p - i\omega x). \tag{3.56}$$

Quantum mechanically this is not exactly H, because x and p do not commute. Using

$$[x, p] = i\hbar, \tag{3.57}$$

one finds

$$\frac{1}{2}(p + i\omega x)(p - i\omega x) = \frac{1}{2}\big(p^2 + \omega^2 x^2 + i\omega[x, p]\big) \tag{3.58}$$

$$= \frac{1}{2}\big(p^2 + \omega^2 x^2 - \hbar\omega\big). \tag{3.59}$$

Hence

$$H = \frac{1}{2}(p + i\omega x)(p - i\omega x) + \frac{1}{2}\hbar\omega. \tag{3.60}$$

The additive constant $\frac{1}{2}\hbar\omega$ is the zero-point energy. In the lecture Susskind temporarily drops it in order to focus on the ladder structure, and then restores it at the end.

Define

$$A_+ = \frac{p + i\omega x}{\sqrt{2\hbar\omega}}, \qquad A_- = \frac{p - i\omega x}{\sqrt{2\hbar\omega}}, \tag{3.61}$$

so that A_+ and A_- are Hermitian conjugates of one another. In standard notation they are $a^\dagger$ and a. The Hamiltonian becomes

$$H = \hbar\omega\left(A_+A_- + \frac{1}{2}\right). \tag{3.62}$$

Now define the number operator

$$N = A_+A_-. \tag{3.63}$$

At this point Susskind asks what Dirac would do. His answer is immediate: commute the factors. A direct calculation gives

$$[A_-, A_+] = 1. \tag{3.64}$$

From this it follows that

$$[N, A_+] = A_+, \qquad [N, A_-] = -A_-. \tag{3.65}$$

The ladder. Suppose

$$N|n\rangle = n|n\rangle. \tag{3.66}$$

Then

$$N(A_+|n\rangle) = (A_+N + [N, A_+])|n\rangle \tag{3.67}$$
$$= (n+1)(A_+|n\rangle), \tag{3.68}$$

so A_+ raises the eigenvalue by one. Similarly,

$$N(A_-|n\rangle) = (n-1)(A_-|n\rangle). \tag{3.69}$$

Thus the oscillator has precisely the same ladder structure that angular momentum had, except that the ladder now lives in energy rather than in m.

What prevents the ladder from descending forever? Here the answer is positivity:

$$\langle\psi|N|\psi\rangle = \langle\psi|A_+A_-|\psi\rangle = \|A_-|\psi\rangle\|^2 \geq 0. \tag{3.70}$$

So the eigenvalues of N are nonnegative. Since raising and lowering change them in integer steps, there must be a lowest one, and it must be

$$n = 0. \tag{3.71}$$

The corresponding ground state obeys

$$A_-|0\rangle = 0. \tag{3.72}$$

With the standard normalization,

$$A_+|n\rangle = \sqrt{n+1}\,|n+1\rangle, \qquad A_-|n\rangle = \sqrt{n}\,|n-1\rangle, \tag{3.73}$$

and so the full energy spectrum is

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots \tag{3.74}$$

This is exactly the spectrum Susskind reaches at the end of the lecture, with the $\frac{1}{2}\hbar\omega$ restored after having been temporarily set aside.

He also remarks, in response to a question, that one does not usually get ladder operators this cleanly. Physics has been lucky here twice: once for angular momentum, and once for the harmonic oscillator. The second luck is especially important because so many systems reduce, near equilibrium, to oscillator behavior.

3.8 Summary

Lecture 3 has a single mathematical instinct running through it. Angular momentum is first reinterpreted as angular dependence of a wavefunction, which leads naturally to the ladder of m-values and to the common multiplet value

$$L^2 = \hbar^2\ell(\ell + 1). \tag{3.75}$$

That algebra then reduces the central-force Schrödinger equation to a radial one-dimensional problem with an effective centrifugal barrier. The resulting spectra are organized by node number and by the $2\ell + 1$ degeneracy of each multiplet, with the Coulomb potential singled out by its special $1, 4, 9, 16, \ldots$ shell pattern before spin is included. The lecture closes by showing that the same algebraic way of thinking works again for the harmonic oscillator: factorize the Hamiltonian, compute a commutator, build a ladder, identify the lowest state, and read off the spectrum.